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ABSTRACT
Smartphones have become ubiquitous in people's lives to an extent that necessitates the investigation of their negative impact. The scarcity of such investigations triggered our interest in examining the relationships among smartphone addiction, anxiety, and family relations. A total of 381 university students participated in an online survey composed of demographic information, the Smartphone Addiction Scale – Short Version, and the Beck Anxiety Inventory. Binary logistic regression analysis and structural equation modelling showed that undergraduate students who had smartphone addiction demonstrated greater odds of having high anxiety compared to those who were not addicted, and students who had high anxiety exhibited greater odds of having clinically significant problems in their family relations. Nevertheless, path analysis showed that anxiety mediated a positive relationship between smartphone addiction and problematic family relations.

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Smartphone addiction; anxiety; family relations; undergraduate students; relationships

1. Introduction

Technological innovations made smartphones the most functional devices encompassing Internet use and a huge variety of apps including texting, gaming, and social networking. Today, smartphones have become an essential part of daily activities to the extent of transforming individuals' lives. Societies are being transformed as well, with the number of smartphone users possibly reaching over 2 billion by the end of 2016 (Statista 2016). The smartphone power and the magnitude of its pervasiveness have made some researchers shift their focus from Internet addiction and problematic mobile phone use to smartphone addiction.

Several studies have been addressing Internet addiction and its association with stress, anxiety, depression, and family relations (Huang et al. 2009; Charlton and Danforth 2010; Hawi 2012; Müller et al. 2014; Kabasakal 2015). These studies have focused mainly on Internet use through desktops or laptops, but the latter are being replaced fast by smartphones. For instance, 15% of American young adults from ages 18–29 years old have become heavily dependent on smartphones for online access (Smith 2015). Of the undergraduate population in the US, 86% owned a smartphone in 2014 compared to 76% in 2013 (Dahlstrom and Bichsel 2014). In Lebanon, 52% of the population owned a smartphone in 2015 (Pew Research Center 2015). The aforementioned data coupled with risk of negative impact of smartphone usage inspired researchers to introduce validated scales that measure smartphone addiction (Carbonell et al. 2012; Kwon et al. 2013b; Lin et al. 2014; Lopez-Fernandez et al. 2014; Kibona and Mgaya 2015; Hawi and Samaha 2016b). In addition, studies examining associations between smartphone use and people's well-being started to emerge. For instance, smartphone addiction has been linked to poor academic performance (Kibona and Mgaya 2015; Hawi and Samaha 2016c), stress, and low satisfaction with life (Lee et al. 2014; Lepp, Barkley, and Karpinski 2014; Samaha and Hawi 2016), and some studies addressed the negative impact of smartphone addiction on people's physical health including hand function and nerves, sleep disturbances and headaches (Canan et al. 2013; Roberts, Yaya, and Manolis 2014; Haug et al. 2015; İnal et al. 2015; Demirci, Demirci, and Akgonul 2016; Hawi and Samaha 2016a).

A study of 269 Taiwanese students showed that anxiety was positively associated with mobile phone addiction (Hong, Chiu, and Huang 2012). A more recent study of 152 Canadian undergraduate university students showed that mobile phone use was predicted by social anxiety (Sapacz, Rockman, and Clark 2016). Another study of 316 Turkish undergraduate students investigated the relationship between smartphone addiction and anxiety and showed that they are positively

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linked (Demirci et al. 2014; Enez Darcin et al. 2016). It is noteworthy that much more research has examined the relationship between Internet Addiction and anxiety, consistently reporting that they were positively correlated (Lee et al. 2012; Dalbudak et al. 2013; Ko et al. 2014).

As for family relations, research addressed its relationship with excessive technology use in children and adults (Yen et al. 2007; Park, Kim, and Cho 2008; Hawi and Rupert 2015; Samaha and Hawi 2017), but not with smartphone addiction. For instance, Internet addiction and problematic Internet use have been linked to family malfunctioning (Li, Garland, and Howard 2014). Also, higher parent-adolescent conflicts and lower family functioning were shown to be predictive of Internet addiction in a sample of Taiwanese high school students with an average age of 15.6 years (Yen et al. 2007). Another study showed that 15–16-year-old boys who were dissatisfied with their families had significantly higher levels of Internet addiction compared to their counterparts (Lam et al. 2009). Other studies demonstrated that problematic Internet use could be predicted in Chinese college students who reported a lack of love from their parents and a poor family atmosphere (Huang et al. 2009). Moreover, a study confirmed that marital conflict between parents can be a risk factor for Internet addiction (Werner and Smith 1992). Another study concluded that there exists a strong correlation between family factors (such as lack of family communication, lack of family cohesion, and family violence) and Internet addiction (Park, Kim, and Cho 2008). A recent study showed that family functioning deteriorated as problematic Internet use increased (Kabasakal 2015). In Lebanon, family values are still very preserved and the invasion of smartphones in the lives of the younger population (Hawi and Samaha 2017) has changed long traditional family habits which may be shaking the long preserved family structures and values. Therefore, it is important to examine if there exists any association between smartphone addiction and family relations.

In our study, we examined smartphone addiction in relation to anxiety and family relations through the validation of the following research hypotheses:

Hypothesis 1: Undergraduate students who have smartphone addiction are more likely to have high anxiety.

Hypothesis 2: Undergraduate students who have high anxiety are more likely to have clinically significant problems in their family relations.

Hypothesis 3: Undergraduate students who have smartphone addiction are more likely to have clinically significant problems in their family relations.

2. Method

2.1. Sampling procedure

Random sampling was used at a private university in Lebanon that adopts the American system of education. A random number generator was used to produce a listing of 700 random numbers without duplicates between 1 and 7000 which was the maximum number of undergraduate students at the University. Students were selected for participation based on these randomly generated numbers. They were provided with a link to access the survey questionnaire and asked to complete it anytime during the spring 2016 semester. A total of 381 survey questionnaires were returned, for a response rate of 54.4%. It is noteworthy that respondents used their smartphones for 'social networking,' 'playing games,' 'texting,' 'reading news,' and 'searching.'

All scales were administered in the English language. A total of 381 undergraduate students responded, filling out the online survey that was approved by the university's research board. A consent form accompanied the request for survey completion, assuring confidentiality of responses and anonymity of respondents. The respondents' average age was 20.84 years old ($SD = 1.92$), with 59.1% males (almost identical to that of the university's student population 59.9%), and ages ranged between 17 and 27 years old (the university's student population ages ranged between 17 and 42 years).

2.2. Instruments

The Smartphone Addiction Scale –Short Version (SAS-SV) (Kwon et al. 2013a) consists of 10 items adopted from the original 33 SAS items (Kwon et al. 2013b). It measures smartphone addiction with a suggested cutoff of 31 for males and 33 for females, with higher score associated with higher risk. SAS-SV showed good psychometric property in our study with a Cronbach's alpha of 0.84. Sample items include: 'Won't be able to stand not having a smartphone' and 'Missing planned work due to smartphone use.'

The Beck Anxiety Inventory (BAI) measures the severity of anxiety by asking respondents about symptoms of anxiety that they had experienced during the past month (Beck et al. 1988). Several studies have examined the BAI on nonclinical samples and showed its strong psychometric properties and validity for the general population (Borden, Peterson, and Jackson 1991; Creamer, Foran, and Bell 1995; Magán, Sanz, and García-Vera 2008). It consists of 21 items rated on a 4-point Likert-type scale, ranging from 'Not at all' (coded 0) to 'Severely' (coded 3). A total score that falls within 0–21, 22–35, or 36–63, indicates low anxiety,

moderate anxiety, or severe anxiety, respectively (Beck and Steer 1990). In our study, the respondents' scores on this scale ranged from 0 to 63. In addition, Cronbach's alpha coefficient was 0.87.

The Index of Family Relations (IFR) is a self-report that measures 'the extent, severity or magnitude of problems that family members have in their relationships with one another' (Hudson 1992). It consists of 25 items rated on a 7-point Likert-type scale, ranging from 'None of the time' (coded 1) to 'All of the time' (coded 7). In our study, the respondents' scores on this scale ranged from 5 to 90. Higher scores on the IFR indicate more severe family problems. In our study, Cronbach's alpha coefficient was .901.

2.3. Statistical analysis

Binary logistic regression on the research variables was performed using IBM SPSS 20.[1] Additionally, structural equation modelling (SEM) was conducted using IBM SPSS Amos Graphics 20 to test the hypothesised models.

Analysis was performed on SAS-SV, BAI, and IFR to test their reliability. The Inter-Item Correlation Matrix of each scale contained no negative values, indicating that the items were measuring the same characteristic. The internal consistency of each one of the scales was measured using Cronbach's alpha (α). The data collected for SAS-SV, BAI, and IFR scales produced internal consistency reliability ($\alpha = 0.846$, $\alpha = 0.904$, and $\alpha = 0.901$, respectively) which were considered very good, excellent, and excellent, respectively.

The Corrected Item-Total Correlation values for SAS-SV ranged from 0.404 (Item 1) to 0.681 (Item 10), indicating that all items measured the same construct as the scale. In addition, none of the values in the column headed 'Alpha if Item Deleted' were higher than 0.846 (α); the values ranged from 0.818 (Item 5) to 0.845 (Item 3), indicative of SAS-SV's homogeneity, and suggesting that no item should be removed from it.

Next, the 10 items of the SAS-SV were subjected to the principal components analysis (PCA) extraction method. The rotation method was Oblimin with Kaiser Normalisation. Prior to performing PCA, the suitability of the data for factor analysis was assessed. Inspection of the correlation matrix revealed the presence of many (80%) coefficients of 0.3 and above. The Kaiser–Meyer–Olkin measure of sampling adequacy (MSA) was 0.890, which exceeded the recommended value of 0.6, and Bartlett's Test of Sphericity reached statistical significance (chi-squared = 918.681, df = 45, $p < .001$), supporting the factorability of the correlation matrix. In fact, PCA revealed the presence of two components with eigenvalues exceeding 1, explaining 42.78% and 11.97% of the variance, respectively. An inspection of the scree plot revealed a clear break after the first component. Using Catell's scree test, it was decided that one component should be retained. This was further supported by the results of the parallel analysis, which showed only one component with an eigenvalue exceeding the corresponding criterion value for a randomly generated data matrix of the same size.

The Corrected Item-Total Correlation values of BAI ranged from 0.406 (Item 18) to 0.684 (Item 8), indicating that all items measured the same construct as the scale. In addition, none of the values in the column headed 'Alpha if Item Deleted' were higher than 0.904 (α); the values ranged from 0.896 (Item 8) to 0.903 (Items 16 and 18), indicative of BAI's homogeneity, and suggesting that no item should be removed from it.

Next, the 21 items of BAI were subjected to the PCA extraction method. The rotation method was Oblimin with Kaiser Normalisation. Prior to performing PCA, the suitability of the data for factor analysis was assessed. Inspection of the correlation matrix revealed the presence of many (70.3%) coefficients of 0.3 and above. The Kaiser–Meyer–Olkin MSAwas 0.894, and Bartlett's Test of Sphericity reached statistical significance (chi-squared = 2421.157, df = 210, $p < .001$), supporting the factorability of the correlation matrix. In fact, PCA revealed the presence of five components with eigenvalues exceeding 1, explaining 34.85%, 7.92%, 6.31%, 4.98%, and 4.81% of the variance, respectively. An inspection of the scree plot revealed a clear break after the first component, and consequently it was decided that one component should be retained. This was further supported by the results of the parallel analysis, which showed only one component with an eigenvalue exceeding the corresponding criterion value for a randomly generated data matrix of the same size.

The Corrected Item-Total Correlation values of IFR ranged from 0.413 (Item 16) to 0.732 (Item 18), indicating that all items measured the same construct as the scale. In addition, none of the values in the column headed 'Alpha if Item Deleted' were higher than 0.901 (α); the values ranged from 0.761 (Item 9) to 0.900 (Items 20), indicative of IFR's homogeneity, and suggesting that no item should be removed from it.

Next, the 25 items of IFR were subjected to the PCA extraction method. The rotation method was Oblimin with Kaiser Normalisation. Prior to performing PCA, the suitability of the data for factor analysis was assessed. Inspection of the correlation matrix revealed the presence of many (83.1%) coefficients of 0.3 and above. The Kaiser–Meyer–Olkin MSA was 0.937, and Bartlett's Test of Sphericity reached statistical significance (chi-squared = 4603.126, df = 300, $p < .001$), supporting the

Table 1. Odds ratios of risk of smartphone addiction, anxiety, and family relations.

	%	Odds Ratio	95% CI		Significant
Anxiety				*High Anxiety?*	
No Risk of SpA	55.2	Reference group			
Risk of SpA	44.8	4.706***	(1.511, 14.659)	Yes	Yes
Family Relations				*Clinically significant problems in family relations?*	
Low Risk of Anxiety	93.9	Reference group			
High Risk of Anxiety	6.1	13.355***	(3.761, 47.428)	Yes	Yes
Family Relations				*Clinically significant problems in family relations?*	
No Risk of SpA	55.2	Reference group			
Risk of SpA	44.8	1.083	(0.660, 1.778)	No	No – CI contains 1

Note: SpA = Smartphone Addiction.
***$p < .001$.

factorability of the correlation matrix. In fact, PCA revealed the presence of four components with eigenvalues exceeding 1, explaining 44.501%, 6.729%, 4.861%, and 4.822% of the variance, respectively. An inspection of the scree plot revealed a clear break after the first component. Using Catell's scree test, it was decided that one component should be retained. This was further supported by the results of the parallel analysis, which showed only one component with an eigenvalue exceeding the corresponding criterion value for a randomly generated data matrix of the same size.

3. Results

Undergraduate students who had smartphone addiction (as measured by SAS-SV) demonstrated greater odds, by 4.706 times ($p = .008$), of having high anxiety (as measured by the Beck Anxiety Inventory) (see Table 1). This result confirmed Hypothesis 1. Undergraduate students who had high anxiety demonstrated greater odds, by 13.355 times ($p = .000$), of having clinically significant problems in their family relations (see Table 1). This result confirmed Hypothesis 2. There was no direct association ($OR = 1.083$, $p = .752$) between smartphone addiction and family relations (as measured by IFR scale). Since the 95% CI of 0.660 to 1.778 spans 1.0, the increased odds (OR = 1.083) of problematic family relations among undergraduate students with smartphone addiction do not reach statistical significance. This is indicated in the aforementioned p-value of .752. In other words, being a student and addicted to smartphone use does not affect the odds of problematic family relations.

This result rejected Hypothesis 3 (see Table 1).

The aforementioned findings triggered the investigation through path analysis of anxiety as a mediator

Figure 1. Conceptual framework using path analysis.

between smartphone addiction and family relations. So, the identification and estimation of the hypothesised path model depicted by Figure 1 was performed in a SEM framework using IBM SPSS Amos Graphics 20. The $\chi 2$ to df ratio (2.211) of the model had a p-value of .137. This ratio indicates an acceptable fit between the model and the sample data (Carmines and Mciver 1981) and that the null hypothesis that the model is correct cannot be rejected (Blunch 2008). The Root Mean Square Error of Approximation (RMSEA) for the model was .064. Thus, the acyclic model is considered an adequate fit because the RMSEA was less than.080 (Browne et al. 1993). Additionally, the normed fit index (NFI), the comparative fit index (CFI), the Tucker–Lewis coefficient (TLI), and the goodness of fit (GFI) were .969, .982, .947, and .995, respectively. All these measures suggested that the model had excellent fit. Additionally, the computed PCLOSE (0.275) was significantly higher than .050, indicating that there was absolutely no evidence to reject the fact that RMSEA was higher than .500. The SRMR (standardised RMR, Root Mean Square Residual) value of .031 indicated excellent fit, as it fell way below the upper limit of.080 (Blunch 2008).

4. Discussion

Smartphones have entered our lives and families in an extensive way. This cross-sectional study is unique in its contribution as it adds to existing work by examining the association of smartphone addiction and family relations. Additionally, the study's data is from Middle East young men and women not often represented in published literature. Furthermore, this study's results are in accordance with those of other studies that reported the relationships between different technology addictions and particularly smartphone addiction and anxiety.

Our results show that the relationship between smartphone addiction and family relations does not exist (see Table 1). This is not in accordance with

other recent studies' results that have shown that smartphone usage by college students has negative impacts on family relationships, $r = -.12$, $N = 643$, $p < .01$ (Chui 2015) and that Problematic Internet Use scores increase as the malfunctioning of university students' families increases, $r = .22$, $N = 663$, $p < .001$ (Kabasakal 2015). Nevertheless, our results show that smartphone addiction is indirectly linked to family relations via anxiety. Our results differ from those of the aforementioned studies probably because of differences in family relations across cultures. In Lebanon, the emphasis on family relations is overly important, so smartphone addiction seems not to have affected these relations 'as of yet.' Nevertheless, the hypothesised model depicting anxiety as a mediator between smartphone addiction and family relations fitted very well with the sample data. According to this model, the SAS-SV score raises the level of anxiety through a direct effect with a regression weight of 0.32, and the direct effect of anxiety on IFR is measured by the regression weight of 0.51.

Smartphone addiction being positively related to anxiety is congruent with another study that included 319 university students from Turkey (Demirci, Akgönül, and Akpinar 2015). For instance, university students with smartphone addiction experienced high levels of anxiety. In other words, although as of yet there is no evidence that smartphone addiction causes anxiety, results indicate that smartphones may make anxiety worse because as smartphone addiction increases, so too does the level of anxiety. Thus, people who use a smartphone with the intention of coping with or easing their anxiety are only aggravating it, which in turn makes family relations more problematic as well. Actually, university students with high levels of anxiety experienced high levels of problematic family relations as well. Furthermore, the hypothesised path model emphasised the bidirectional nature of relationships whereby variables have reciprocal influences. It was suggested that Internet addiction and psychiatric symptoms can interact with and precipitate each other (Beranuy et al. 2009). In our context, the higher the smartphone addiction is, the higher the level of anxiety would be, and the higher the level of anxiety is, the higher smartphone addiction would be, which leads to a vicious circle. In other words, a stressor that raises the level of anxiety might also increase smartphone addiction. In our context, a high level of problematic family relations is a stressor. Another example of a stressor is perceived stress (Samaha and Hawi 2016). The literature states that anxiety is a reaction to stress that is caused by an existing stress-causing factor or stressor (Magalhaes et al. 2010). Unfortunately, anxiety continues after the stressor is gone by increasing in importance to the extent that it makes a person feel afraid.

In short, a factor that places a student with smartphone addiction increases the odds by about five times of having high anxiety, which propels a student into a problematic mental status that is characterised by smartphone addiction, a high level of anxiety, and a high level of problematic family relations. Additionally, students experiencing clinically significant problems in their family relations were more likely to experience high anxiety, which made them more likely to be prone to smartphone addiction. This pattern is a novel contribution to the literature and is worth further investigation.

Given that the smartphone era is relatively new, the availability of prior research investigating the relationship between smartphone addiction, anxiety, and family relations is very limited. Nevertheless, this study's results uncovered links between them, forming more than one opportunity thread for further investigations.

4.1. Limitations and future work

Although the SAS-SV was originally developed for adolescents, it was used and validated for older population. For instance it was validated on Turkish university students (Noyan et al. 2015), on Swiss non-vocal students including young adults (18+) (Haug et al. 2015) and on a population of over 18 of Spanish and Belgian adults (Lopez-Fernandez 2017). Longitudinal studies are needed to confirm the results of our cross-sectional study with other sampling methods such as stratified random sampling. Another limitation of the study is the sample size which should have been bigger and encompassing different smartphone user groups and not just university students. Also, the self-rated nature of the scales in this study might make answers biased based on social desirability. Furthermore, more studies should be conducted in different cultural contexts, especially because family values and relations differ in different parts of the world.

5. Conclusion

The results of our study suggest the existence of a relationship between smartphone addiction and anxiety and indirect relationship between smartphone addiction and family relations. They add to the growing literature of researchers calling to consider seriously all kind of behavioural technology-related addictions including smartphone addiction. These findings form a conceptual framework that can mediate the development of intervention programs that encourage and promote

awareness campaigns on smartphone addiction. In this fast-growing digital age, there is an urgent need to develop and implement different strategies that focus on educating parents and children about the detrimental effects of smartphone addiction to prevent situations of anxiety and deterioration in family relations.

Note

1. SPSS 20.0 software proprietary of IBM, New York, US.

Disclosure statement

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